

Exercise JA

Q1. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is

- (A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}

Ans. D

Sol.

$$|Q| = \begin{vmatrix} 2^2 a_{11} & 2^3 a_{12} & 2^4 a_{13} \\ 2^3 a_{21} & 2^4 a_{22} & 2^5 a_{23} \\ 2^4 a_{31} & 2^5 a_{32} & 2^6 a_{33} \end{vmatrix}$$

$$\Rightarrow |Q| = 2^2 \cdot 2^3 \cdot 2^4 \cdot \begin{vmatrix} a_{11} & 2a_{12} & 2^2 a_{13} \\ a_{21} & 2a_{22} & 2^2 a_{23} \\ a_{31} & 2a_{32} & 2^2 a_{33} \end{vmatrix}$$

$$= 2^2 \cdot 2^3 \cdot 2^4 |P| \cdot 2^3$$

$$= 2^2 \cdot 2^3 \cdot 2^4 \cdot 2 \cdot 2^3 = 2^{13}$$

Q2. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the

3×3 identity matrix, then there exists a column matrix $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

- (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$

Ans. D

Sol.

$$P^T = 2P + I$$

$$\Rightarrow P = 2P^T + I$$

$$\Rightarrow P = 2(2P + I) + I$$

$$\Rightarrow P = 4P + 3I$$

$$\Rightarrow P = -I$$

$$\Rightarrow PX = -X$$

Q3. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are)

- (A) -2 (B) -1 (C) 1 (D) 2

Ans. D, A

Sol.
$$|\text{adj } P| = \begin{vmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{vmatrix}$$

$$\Rightarrow |P|^2 = 4 \quad \Rightarrow |P| = 2$$

Q4. For 3×3 matrices M and N, which of the following statment(s) is (are) NOT correct ?

- (A) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
- (B) $M N - N M$ is skew symmetric for all symmetric matrices M and N
- (C) $M N$ is symmetric for all symmetric matrices M and N
- (D) $(\text{adj } M) (\text{adj } N) = \text{adj}(M N)$ for all invertible matrices M and N

Ans. D, C

Sol. (C,D) Ans.

(A) $(N^T M N)^T = N^T M^T N$ is symmetric if M is symmetric and skew-symmetric if M is skew-symmetric.

(B) $(M N - N M)^T = (M N)^T - (N M)^T = N M - M N = -(M N - N M)$ skew symmetric

(C) $(M N)^T = N^T M^T = N M \neq M N$ hence NOT correct

(D) Standard result is $\text{adj}(M N) = (\text{adj } N) (\text{adj } M) \neq (\text{adj } M) (\text{adj } N)$

Q5. Let M and N be two 3×3 matrices such that $M N = N M$. Further, if $M \neq N^2$ and $M^2 = N^4$, then

- (A) determinant of $(M^2 + M N^2)$ is 0
- (B) there is a 3×3 non-zero matrix U such that $(M^2 + M N^2) U$ is the zero matrix
- (C) determinant of $(M^2 + M N^2) \geq 1$
- (D) for a 3×3 matrix U, if $(M^2 + M N^2) U$ equals the zero matrix then U is the zero matrix

Ans. A, B

Sol. $M N = N M$ & $M^2 - N^4 = 0$

$$\begin{array}{c} (M - N^2)(M + N^2) = 0 \\ \hline \begin{array}{ccc} M - N^2 = 0 & M + N^2 = 0 & |M + N^2| = 0 \\ \text{Not Possible} & M - N^2 \neq 0 & |M - N^2| = 0 \end{array} \\ \hline \text{In any case } |M + N^2| = 0 \end{array}$$

(A) $|M^2 + MN^2| = |M| |M + N^2| = 0$ (A) is correct

(B) If $|A| = 0$ then $AU = 0$ will have ∞ solution. Thus $(M^2 + MN^2)U = 0$ will have many 'U' (B) is correct

(C) Obvious wrong.

(D) If $AX = 0$ & $|A| = 0$ then X can be non zero. (D) is wrong

Q6. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if

(A) the first column of M is the transpose of the second row of M

(B) the second row of M is the transpose of the column of M

(C) M is a diagonal matrix with nonzero entries in the main diagonal

(D) the product of entries in the main diagonal of M is not the square of an integer

Ans. D, C

Sol. $M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

(A) $\begin{bmatrix} a \\ b \end{bmatrix}$ & $[b \ c]$ are transpose.

So $\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ c \end{bmatrix} \Rightarrow$ is given $a = b = c$

$M = \begin{bmatrix} a & a \\ a & a \end{bmatrix} \Rightarrow |M| = 0$ A is wrong

(B) $[b \ c]$ & are transpose.

So $a = b = c$ B is wrong

(C) $M = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \Rightarrow |M| = ac \neq 0$ C is correct

(D) $M =$ given $ac \neq \lambda^2$. D is correct

(C, D) are correct.

Q7. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric ?

(A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

Ans. D, C

Sol. $(X^4Z^3 - Z^3X^4)^T = (X^4Z^3)^T (Z^3X^4)^T$
 $= (Z^T)^3 (X^T)^4 - (X^T)^4 (Z^T)^3$

$$= Z^3 X^4 - X^4 Z^3$$

$$= -(X^4 Z^3 - Z^3 X^4)$$

$$(D) (X^{23} + Y^{23})^T = -X^{23} - Y^{23} \Rightarrow X^{23} + Y^{23} \text{ is skew-symmetric}$$

Q8.

Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$,

where $k \in \mathbb{R}$, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then

(A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$ (C) $\det(P \operatorname{adj}(Q)) = 2^9$

(D) $\det(Q \operatorname{adj}(P)) = 2^{13}$

Ans. C, B

Sol.

$$P \left(\frac{Q}{K} \right) = I$$

$$\Rightarrow P^{-1} = \left(\frac{Q}{K} \right)$$

$$\Rightarrow (P^{-1})_{23} = \frac{q_{23}}{K} = -\frac{1}{8}$$

$$\Rightarrow \frac{-(3\alpha + 24)}{12\alpha + 20} = -\frac{1}{8} \Rightarrow -1$$

$$\text{Hence } |P| = 12\alpha + 20 = 8$$

$$\text{Again } |P| \left| \frac{Q}{K} \right| = |I| \Rightarrow \frac{8|Q|}{K^3} = 1$$

$$\Rightarrow |Q| = \frac{K^3}{8}$$

$$\Rightarrow \frac{K^3}{8} = \frac{K^2}{2} \Rightarrow K = 4 \Rightarrow |Q| = 8$$

$$(C) \det(P \operatorname{adj}(Q)) = \det(P) \cdot \det(\operatorname{adj}(Q))$$

$$= \det(P) \cdot (\det(Q))^2 = 8 \times 8^2 = 2^9$$

$$(D) \det(Q \operatorname{adj}(P)) = \det(Q) \cdot (\det P)^2 = 8 \times 8^2 = 2^9$$

Q9.

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such

that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals

(A) 52 (B) 103 (C) 201 (D) 205

Ans. B

Sol.
$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} = I + A$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^n = 0 \quad \forall n \geq 3$$

Given : $P^{50} = Q + I$

$$I + Q = (I + A)^{50} = I + 50A + \frac{50 \times 49}{2} A^2 + 0$$

$$\Rightarrow Q = 50 \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} + \frac{50 \times 49}{2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$q_{31} = 50 \times 16 + \left(\frac{50 \times 49}{2} \right) 16 = 200 \times 102$$

$$q_{32} = 50 \times 4 \times 200, \quad q_{21} = 50 \times 4 = 200$$

$$\frac{q_{31} + q_{32}}{q_{21}} = 103$$

Q10. Which of the following is(are) NOT the square of a 3×3 matrix with real entries?

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Ans. D, B

Sol. Let A be a 3×3 matrix with real entries, and let $B = A^2$. Taking determinant:

$\det(B) = \det(A^2) = (\det(A))^2 \geq 0$. So if $\det(B) < 0$, then B cannot be a square of a real matrix.

Option (A): $B = I$, $\det(B) = 1 \geq 0$. Can choose $A = I$ (real). So it is a square.

Option (B): $\det(B) = (-1)(-1)(-1) = -1 < 0$. Thus cannot be a square.

Option (C): $\det(B) = 1 \cdot (-1) \cdot (-1) = 1 \geq 0$. Can construct $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ (real). Then

$A^2 = B$. So it is a square.

Option (D): $\det(B) = 1 \cdot 1 \cdot (-1) = -1 < 0$. Thus cannot be a square.

Therefore, the correct options are B and D.

Q11. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5?

(A) 162 (B) 135 (C) 126 (D) 198

Ans. D

Sol. $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

$$a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 5$$

Case - I : Five (1's) and four (0's)

$${}^9C_5 = 126$$

Case - II : One (2) and one (1)

$${}^9C_2 \times 2! = 72$$

$$\therefore \text{Total} = 198$$

Q12. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in R$ and the system of equations (in real variables)

$$-x + 2y + 5z = b_1$$

$$2x - 4y + 3z = b_2$$

$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables)

has (have) at least one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

(A) $x + 2y + 3z = b_1, 4y + 5z = b_2$ and $x + 2y + 6z = b_3$

(B) $x + y + 3z = b_1, 5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$

(C) $-x + 2y - 5z = b_1, 2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$

(D) $x + 2y + 5z = b_1, 2x + 3z = b_2$ and $x + 4y - 5z = b_3$

Ans. D, A

Sol. $D = 0$ so for at least one solutions $D_1 = D_2 = D_3 = 0 \Rightarrow b_1 + 7b_2 = 13b_3 \dots\dots\dots(i)$

option (A) $D = 0 \Rightarrow$ unique solution $\Rightarrow$ option (A) is correct

option (D) $D = 0 \Rightarrow$ unique solution $\Rightarrow$ option (B) is correct

option (C) $D = 0 \Rightarrow$ equations are $x - 2y + 5z = -b_1$

$$x - 2y + 5z = \frac{b_2}{2}$$

$$x - 2y + 5z = b_2$$

There planes are parallel so they must be coincident

$\Rightarrow$ which satisfies equation (1) for all $b_1, b_2, b_3 \Rightarrow$ option (C) is correct.

option (B) $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 5 & 2 & 2 \\ 2 & 1 & 1 \end{vmatrix} = 0$. Also $D_1 = 0$

For infinite solutions, D_2 and D_3 must be 0

$$\Rightarrow \begin{vmatrix} 1 & b_1 & 1 \\ 5 & b_2 & 2 \\ 2 & b_3 & 1 \end{vmatrix} = 0 \Rightarrow -b_1 - b_2 + 3b_3 = 0 \text{ which does not satisfy (i) for all } b_1, b_2, b_3$$

so option(s)

wrong

Q13. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is _____ .

Ans. 4

Sol. $\det(P) = \begin{vmatrix} a_1 & b_2 & c_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1) \leq 6$

value can be 6 only if $a_1 = 1, a_2 = -1, a_3 = 1, b_2c_3 = b_1c_3 = 1, b_3c_2 = b_3c_1 = -1$

$$\Rightarrow (b_2c_3)(b_3c_1)(b_1c_2) = -1 \quad \& \quad (b_1c_3)(b_3c_2)(b_2c_1) = 1$$

i.e. $b_1b_2b_3c_1c_2c_3 = 1$ and -1

hence not possible

Similar contradiction occurs when $a_1 = 1, a_2 = 1, a_3 = 1, b_2c_2 = b_3c_1 = b_1c_2 = 1, b_3c_2 =$

$$b_1 c_3 = b_1 c_2 = -1$$

Now for value to be 5 one the terms must be zero but that will make 2 terms zero which means answer cannot be 5

$$\text{Now } (P) = \begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & -2 & 1 \end{vmatrix} = 4 \text{ Hence max value} = 4$$

Q14. Let $= \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} \alpha I + \beta M^{-1}$

where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers and I is the 2×2 identity matrix. If α^* is the minimum of the set $\{\alpha(\theta): \theta \in [0, 2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta): \theta \in [0, 2\pi)\}$

then the value of $\alpha^* + \beta^*$

(A) $-\frac{31}{16}$ (B) $-\frac{29}{16}$ (C) $-\frac{37}{16}$ (D) $-\frac{17}{16}$

Ans. B

Sol. $M = \alpha I + \beta M^{-1}$

$$M^2 = \alpha M + \beta I$$

$$M^2 - \alpha M - \beta I = 0$$

$$\alpha = \text{tr}(M) = \sin^4 \theta + \cos^4 \theta$$

$$\alpha = 1 - \frac{1}{2} \sin^2 2\theta$$

$$(\alpha)_{\min} = 1 - \frac{1}{2} = \frac{1}{2} \text{ when } \sin^2 2\theta = 1$$

$$-\beta = |M|$$

$$\beta = -|M|$$

$$= -(\sin^4 \theta \cos^4 \theta + (1 + \sin^2 \theta)(1 + \cos^2 \theta)) = -(\sin^4 \theta \cos^4 \theta + \sin^2 \theta + \cos^4 \theta + 2)$$

$$= -\left(\left(\sin^2 \theta \cos^2 \theta + \frac{1}{2}\right)^2 + \frac{7}{4}\right) = -\left(\left(\frac{\sin^2 2\theta}{4} + \frac{1}{2}\right)^2 + \frac{7}{4}\right)$$

$$(\beta)_{\min} = \frac{-37}{16} \text{ when } \sin^2 2\theta = 1$$

$$(\alpha)_{\min} + (\beta)_{\min} = \frac{-29}{16}$$

Q15. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj } M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$

Where a and b are real numbers, which of the following options is/are correct

(A) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

$$(B) \text{ If } M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ then } \alpha - \beta + \gamma = 3 / \text{ if } M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ then } \alpha - \beta + \gamma = 3$$

$$(C) a + b = 3 \quad (D) \det(\text{adj} M^2) = 81$$

Ans. A, B, C

Sol.

$$M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$$

$$\text{adj}(M) = \begin{bmatrix} 2-3b & \dots & \dots \\ \dots & -3a & \dots \\ \dots & \dots & \dots \end{bmatrix} \quad 2-3b = -13b = 3b = 1 \quad -3a = -6a = 2$$

$$M = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \Rightarrow |M| = -2$$

$$(1) \text{adj}(M^{-1}) + (\text{adj } M)^{-1} = -M$$

$$(\text{adj } M)^{-1} + (\text{adj } M)^{-1} = -M$$

$$2(\text{adj}(M))^{-1} = -M$$

$$2(\text{adj } M)^{-1} \text{adj } M = -M \text{adj } M$$

$$2I = -|M| I$$

$$2I = 2I$$

$$(2) \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$M^{-1} = \frac{\text{adj}(M)}{|M|} = -\frac{1}{2} \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= -\frac{1}{2} \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} \quad \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \Rightarrow \alpha = 1, \beta = -1, \gamma = 1$$

$$(3) a + b = 1 + 2 = 3$$

$$(4) |\text{adj}(M^2)| = |M^2|^2 = |M^4| = 16$$

Q16. Let $x \in \mathbb{R}$ and let

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \text{ and } R = PQP^{-1}$$

Then which of the following options is are/correct

$$(A) \text{ For } x = 1 \text{ there exists a unit vector } \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k} \text{ for which } R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(B) \det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8, \text{ for all } x \in \mathbb{R}$$

$$(C) \text{ For } x = 0, \text{ if } R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} \text{ then } a + b = 5$$

$$(D) \text{ There exists a real number } x \text{ such that } PQ = QP$$

Ans. C, B

Sol. Option (2) $PQP^{-1} = R$

$$PQ = RP$$

$$\det P \det Q = \det R \det P$$

$$\det R = \det Q$$

$$\det Q = 48 - 4x^2$$

$$2^{\text{nd}} \text{ option} \Rightarrow 40 - x^2 + 8$$

$$R = PQP^{-1}$$

$$\text{Option (3)} R = \frac{1}{6} \begin{bmatrix} 6x + 12 & 3x + 6 & 4 - 10x \\ 12 & 24 & 8 - 4x \\ 18x & 0 & 36 - 6x \end{bmatrix}$$

$$\text{if } x = 0$$

$$R = \frac{1}{6} \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = \begin{bmatrix} 4a + \frac{4}{3}b \\ 6b \end{bmatrix} \Rightarrow a = 2 \text{ and } b = 3$$

$$a + b = 5$$

Option (4) $PQ = QP$ not possible

Option (1) if we put $x = 1$ then

$$R = \frac{1}{6} \begin{bmatrix} 18 & 9 & -6 \\ 12 & 24 & 4 \\ 18 & 0 & 30 \end{bmatrix}$$

$$|R| \neq 0$$

$\Rightarrow$ system will have unique solution

$\Rightarrow$ only $\alpha = \beta = \gamma = 0$ will satisfy the system

Q17. Let

$$P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$, P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\text{and } X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} P_k^T$$

where P_k^T denotes the transpose of the matrix P_k then which of the following options is/are correct

$$(A) \text{ The sum of diagonal entries of } X \text{ is } 18 \quad (B) \text{ If } X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ then } a = 30$$

$$(C) X \text{ is a symmetric matrix} \quad (D) X - 30I \text{ is an invertible matrix}$$

Ans. C, B, A

$$\text{Sol. } P^1 = P_1^T = P_1^{-1}$$

$$P_2 = P_2^T = P_2^{-1}$$

$$P_6 = P_6^T = P_6^{-1}$$

$$\text{and } A^T = A$$

$$(A + B)^T = A^T + B^T$$

$$X^T = (P_1 A P_1^T + \dots P_6 A P_6^T)^T$$

$$= X \text{ so } X \text{ is symmetry}$$

$$\text{trace } (2 + 0 + 1) + (2 + 0 + 1) \dots 6 \text{ time} = 18$$

$$\text{Let } B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$XB = P_1 A P_1^T B + \dots P_6 A_6^T B$$

$$XB = (P_1 + P_2 + P_3 + \dots + P_6) \begin{bmatrix} 6 \\ 3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \times 2 & + 3 \times 2 & + 6 \times 2 \\ 6 \times 2 & + 3 \times 2 & + 6 \times 2 \\ 6 \times 2 & + 3 \times 2 & + 6 \times 2 \end{bmatrix} = \begin{bmatrix} 30 \\ 30 \\ 30 \end{bmatrix} = 300 \Rightarrow \alpha = 30$$

$$\text{Since } x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 30 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow (x - 30)B = 0 \text{ has non trivial solution } B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow |x - 30| = 0$$

Q18. Suppose

$$\det \begin{bmatrix} \sum_{k=0}^n k & \sum_{k=0}^n {}^n C_k k^2 \\ \sum_{k=0}^n {}^n C_k k & \sum_{k=0}^n {}^n C_k 3^k \end{bmatrix} = 0$$

holds for some positive integer n then $\sum_{k=0}^n \frac{{}^n C_k}{k+1}$ equals

Ans. 6.2

$$\text{Sol. } \begin{bmatrix} \frac{n(n+1)}{2} & n(n+1)2^{n-2} \\ n2^{n-1} & 4^n \end{bmatrix} = 0$$

$$n = 4$$

$$\text{put } n = 4 \text{ In } \sum_{k=0}^n \frac{{}^n C_k}{k+1}$$

$$\sum_{k=0}^4 \frac{{}^4 C_k}{k+1} = \frac{2^5 - 1}{5} = \frac{31}{5}$$

$$= 6.2$$

Q19. Let M be a 3×3 invertible matrix with real entries and let I denote the 3×3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statements is / are ALWAYS TRUE ?

(A) $M = I$ (B) $\det M = 1$ (C) $M^2 = I$ (D) $(\text{adj } M)^2 = I$

Ans. B, C, D

Sol. $M^{-1} = \text{adj}(\text{adj}(M))$
 $M^{-1} = |M|^{-1} M$
 $I = |M| M^2 \dots\dots\dots(1)$
 $|I| = | |M| M^2 |$
 $I = |M|^3 |M|^2$
 $|M|^5 = 1 \Rightarrow |M| = 1 \dots\dots\dots(2)$
 $\Rightarrow$ Put in (1)
 $M^2 = I \Rightarrow M^{-1} = M$
 $\frac{\text{adj}(M)}{|M|} = M$
 $\text{adj}(M) = M$
 $\Rightarrow (\text{adj}(M))^2 = I$

Q20. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is -18 , then the value of the determinant of A is _____

Ans. 5

Sol. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $a + d = 3$
 $d = 3 - a$

$$A = \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix}$$

$$|A| = a(3-a) - bc$$

$$|A| = 3a - a^2 - bc$$

$$A^2 = \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix} \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix} = \begin{bmatrix} a^2 + bc & 3b \\ 3c & (3-a)^2 + bc \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^2+bc & 3b \\ 3c & (3-a)^2+bc \end{bmatrix} \begin{bmatrix} a & b \\ c & 3-a \end{bmatrix}$$

$$\text{tr}(A^3) = a^3 + abc + 3bc + 3bc + (3-a)[(3-a)^2 + bc] = -18$$

$$\text{tr}(A^3) = a^3 + abc + 6bc + (3-a)^3 + 3bc - abc = -18$$

$$\text{tr}(A^3) = a^3 + 9bc + (3-a)^3 = -18$$

$$\text{tr}(A^3) = a^3 + 9bc + 27 - a^3 - 9a(3-a) = -18$$

$$9bc + 27 - 27a + 9a^2 = -18$$

$$bc + 3 - 3a + a^2 = -2$$

$$5 = 3a - a^2 - bc$$

hence

$$|A| = 3a - a^2 - bc$$

$$|A| = 5$$

Q21. For any 3×3 matrix M, let $|M|$ denote the determinant of M. Let

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a nonsingular matrix of order 3×3 , then which of the following statements is (are) TRUE ?

$$(A) F = PEP \text{ and } P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (B) |EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$$

$$(C) |(EF)^3| > |EF|^2$$

$$(D) \text{ Sum of the diagonal entries of } P^{-1}EP + F \text{ is equal to the sum of diagonal entries of } E + P^{-1}FP$$

Ans. A, D

Sol.

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}; P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}; F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

$|Q| \neq 0$, Q is square matrix of order 3

$$(A) PE = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 13 & 18 \\ 2 & 3 & 4 \end{bmatrix}$$

$$PEP = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 13 & 18 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix} = F$$

$$P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ (A) is Correct

(B) $F = PEP$ & $P^2 = I$ (from (A))

$$PF = P^2EP$$

$$PF = IEP$$

$$PF = EP \text{ -----(1)}$$

$$\text{LHS} = |EQ + PFQ^{-1}|$$

$$= |EQ + EPQ^{-1}|$$

$$= |E||Q + PQ^{-1}|$$

$$= 0$$

$$|E| = 0$$

$$(C) |EF| = |E||F| = 0$$

$\Rightarrow$ (C) is Incorrect

LHS :

$$\text{RHS} = |EQ| + |PFQ^{-1}|$$

$$= |EQ| + |EPQ^{-1}|$$

$$= |E||Q| + |E||PQ^{-1}|$$

$$= 0$$

$$\begin{aligned}
 & \text{(D) } \text{tr}(P^{-1}EP + F) \\
 &= \text{tr}(P^{-1}PF + F) \quad (\text{Using (1)}) \\
 &= \text{tr}(F + F) \\
 &= \text{tr}(2F) = 2 \text{tr}(F) = 2(1 + 18 + 3) = 44
 \end{aligned}$$

RHS :

$$\begin{aligned}
 & \text{tr}(E + P^{-1}FP) \\
 &= \text{tr}(E + P^{-1}EP) \quad (\text{Using (1)})
 \end{aligned}$$

$$\text{Now, } P^2 = I \Rightarrow P = P^{-1}$$

$$\begin{aligned}
 &= \text{tr}(E + PEP) \\
 &= \text{tr}(E) + \text{tr}(PEP) \\
 &= \text{tr}(E) + \text{tr}(F) \\
 &= (1 + 3 + 18) + (1 + 18 + 3) \\
 &= 44
 \end{aligned}$$

$\Rightarrow$ (D) is correct

Q22. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) TRUE ?

$$(A) |FE| = |I - FE| |FGE| \quad (B) (I - FE)(I + FGE) = I \quad (C) EFG = GEF$$

$$(D) (I - FE)(I - FGE) = I$$

Ans. A, B, C

$$\text{Sol. } G = (I - EF)^{-1}$$

$$\Rightarrow I - EF = G^{-1}$$

$$\text{Now, } G.G^{-1} = G^{-1}.G$$

$$G(I - EF) = (I - EF).G$$

$$G - GEF = G - EFG$$

$$GEF = EFG \quad \dots\dots\dots(1)$$

(C) is correct

Now,

$$\begin{aligned}
 & (A) (I - FE)(FGE) \\
 &= FGE - FEFGE \quad \dots\dots\dots(2)
 \end{aligned}$$

$$I - EF = G^{-1}$$

$$\Rightarrow G(I - EF) = G.G^{-1}$$

$$\Rightarrow G - GEF = I$$

$$\Rightarrow G - EFG = I \quad [\text{From (1)}]$$

$$\Rightarrow G - I = EFG \quad \dots\dots\dots(3)$$

Put $G - I = EFG$ in (2), we get

$$FGE - FEFGE$$

$$\begin{aligned}
&= FGE - F(G-I)E \\
&= FGE - FGE + FE \\
&= FE
\end{aligned}$$

$$\Rightarrow (I-FE)(FGE) = FE$$

$$\Rightarrow |I-FE| |FGE| = |FE|$$

$\Rightarrow$ (A) is correct

$$(B) (I-FE)(I+FGE)$$

$$\begin{aligned}
&= I + FGE - FE - FE FGE \\
&= I + FGE - FE - F(G-I)E \quad (\text{from (3)}) \\
&= I + FGE - FE - FGE + FE \\
&= I
\end{aligned}$$

$\Rightarrow$ (B) is correct

$$(C) (I-FE)(I-FGE)$$

$$\begin{aligned}
&= I - FGE - FE - FE FGE \\
&= I - FGE - FE + F(G-I)E \quad (\text{from (3)}) \\
&= I - FGE - FE + FGE - FE \\
&= I - 2FE
\end{aligned}$$

$\Rightarrow$ (D) is incorrect

Q23. Let β be a real number. Consider the matrix $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$. If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____.

Ans. 3

Sol. $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, so

$$|A^7 - (\beta - 1)A^6 - \beta A^5| = 0$$

$$|A|^5 |(A^2 - (\beta - 1)A - \beta I)| = 0$$

$$|A|^5 |A^2 - (\beta - 1)A - \beta I| = 0$$

$$|A|^5 |A^2 - \beta A + A - \beta I| = 0$$

$$|A|^5 |(A + I)(A - \beta I)| = 0$$

here $|A| = -1$

$$|A + I| = \begin{vmatrix} \beta + 1 & 0 & 1 \\ 2 & 2 & -2 \\ 3 & 1 & -1 \end{vmatrix} = 0 + 1(2 - 6) = -4$$

$$\text{So } |A - \beta I| = \begin{vmatrix} 0 & 0 & 1 \\ 2 & 1 - \beta & -2 \\ 3 & 1 & -2 - \beta \end{vmatrix}$$

$$\Rightarrow 2 - 3 + 3\beta = 0$$

$$\Rightarrow 3\beta = 1$$

$$\text{So } 9\beta = 3$$

Q24. If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

- (A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$ (B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$ (C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$
 (D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

Ans. A

Sol. Let $M = B + I$

$$B = \begin{bmatrix} \frac{3}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{3}{2} \end{bmatrix}, B^2 = 0 \text{ and hence } B^3, B^4, \dots = 0$$

$$M^{2022} = (B + I)^{2022} = I + 2022 B = \begin{bmatrix} 3034 & 3033 \\ -3033 & -3032 \end{bmatrix}$$

Q25. Let $M = (a_{ij})$, $i, j \in \{1, 2, 3\}$, be the 3×3 matrix such that $a_{ij} = 1$ if $j + 1$ is divisible by i , otherwise $a_{ij} = 0$. Then which of the following statements is(are) true?

(A) M is invertible

(B) There exists a non zero column matrix $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ such that $M \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

(C) The set $\{X \in \mathbb{R}^3 : MX = 0\} \neq \{0\}$ where $0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

The matrix $(M - 2I)$ is invertible, where I is the 3×3 identity matrix

(D) Let $M = (a_{ij})$, $i, j \in \{1, 2, 3\}$, be the 3×3 matrix such that $a_{ij} = 1$ if $j + 1$ is divisible by i , otherwise $a_{ij} = 0$. Then which of the following statements is(are) true?

Ans. B, C

Sol.
$$M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$|M| = -1 + 1 = 0 \Rightarrow M$ is singular so non-invertible

$$(B) M \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} -a_1 \\ -a_2 \\ -a_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} -a_1 \\ -a_2 \\ -a_3 \end{bmatrix}$$

$$\left. \begin{array}{l} a_1 + a_2 + a_3 = -a_1 \\ a_1 + a_3 = -a_2 \\ a_2 = -a_3 \end{array} \right\} \Rightarrow a_1 = 0 \text{ and } a_2 + a_3 = 0 \text{ infinite solutions exists [B] is correct.}$$

Option (D)

$$M - 2I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

$|M - 2I| = 0 \Rightarrow [D]$ is wrong

Option (C) :

$$MX = 0 \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + y + z = 0$$

$$x + z = 0$$

$$y = 0$$

$\therefore$ Infinite solution

[C] is correct

Q26. Let $R = \left\{ \begin{pmatrix} a & 3 & b \\ c & 2 & d \\ 0 & 5 & 0 \end{pmatrix} : a, b, c, d \in \{0, 3, 5, 7, 11, 13, 17, 19\} \right\}$ Then the number of invertible matrices in R is

Ans. 3780

Sol. $|R| = -5(ad - bc)$

$$\begin{aligned} \text{invertible matrices} &= \text{Total} - (\text{non-invertible}) \\ &= 8^4 - (\text{when } ad - bc = 0) \end{aligned}$$

$$ad = bc$$

Case - I

Non zero = non zero

(I) all are same

7 cases

(II) Two alike, two other alike

Ex : (a = b and c = d)

$$= {}^7C_2 \cdot {}^2C_1 \cdot {}^2C_1 = 84 \text{ cases}$$

Case - II

zero = zero

(i) all are zero

zero

$$= 1 \text{ case}$$

(ii) any three are zero

$${}^7C_1 \cdot {}^4C_1 = 28$$

(iii) one of (a, d) and one of (b, c) is

$$\begin{aligned} &= {}^2C_1 \cdot {}^2C_1 \cdot {}^7C_1 \cdot {}^7C_1 \\ &= 196 \end{aligned}$$

So total invertible matrices

$$= 8^4 - (\text{case 1}) - (\text{case 2})$$

$$= 4096 - (7 + 84) - (1 + 28 + 196)$$

$$= 3780$$

Ans. 3780